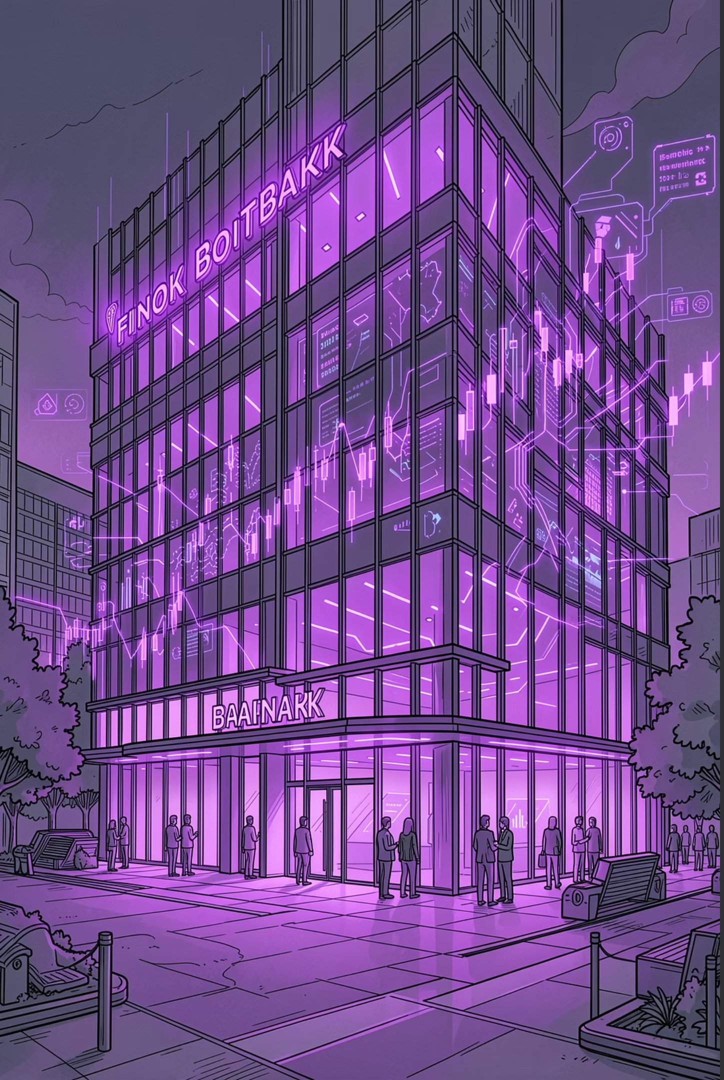
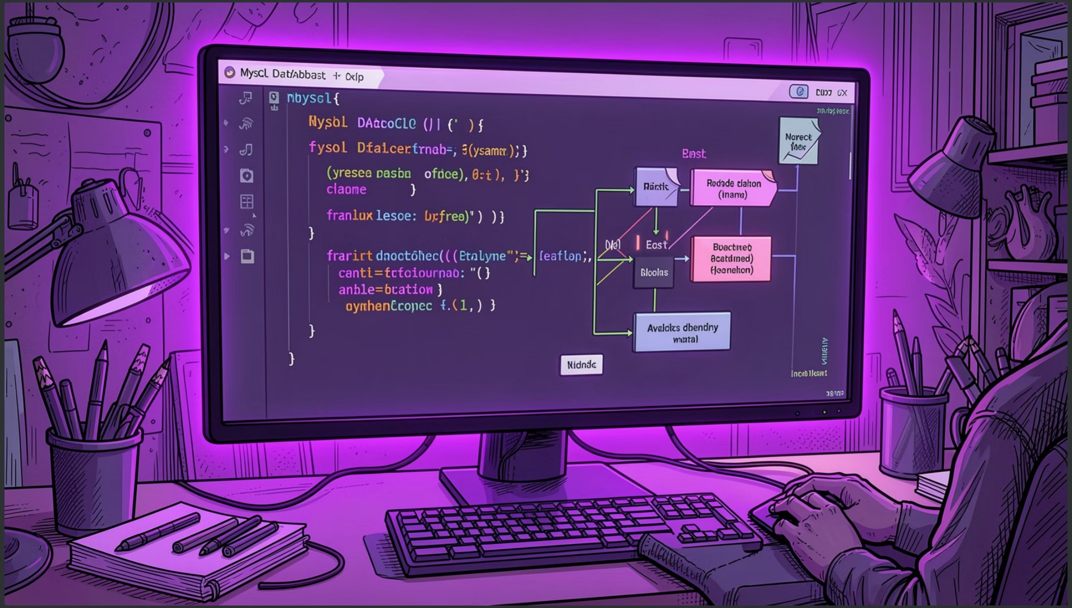


🏦 Infinite Wallet Bank

♾️ Infinite Trust, Infinite Growth 📈

A Python + MySQL banking project featuring account creation, transactions, money transfers, and an admin panel.

Project Overview



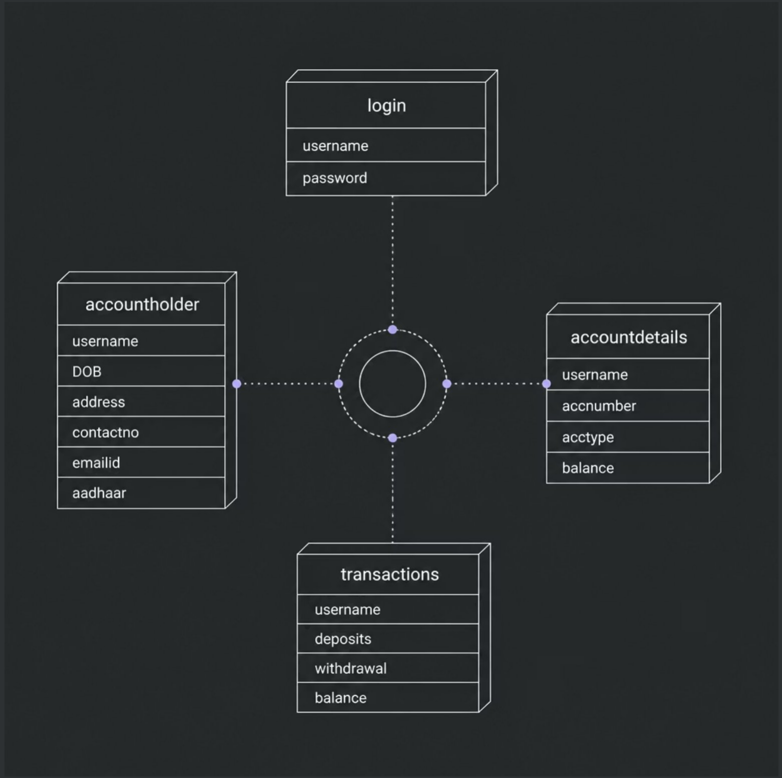
Tech Stack

What It Does

Full-featured online banking portal with customer login, admin panel, real-time balance tracking, and secure money transfers — all powered by a MySQL database.

Python	MySQL	PrettyTable
Core logic & CLI	Database backend	Formatted output

Database Architecture



Relational Schema

Four tables linked by **username** as the primary key, ensuring data consistency across login credentials, personal details, account info, and transaction history.

- **login** — Authentication
- **accountholder** — KYC data
- **accountdetails** — Account & balance
- **transactions** — Deposit/withdrawal log

Create Account

01

Login Credentials

Enter username & password, stored in `login` table

02

Personal Details

DOB, address, contact, email, and Aadhaar number

03

Account Type

Choose from Savings, Current, FD, RD, NRI, Salary, Demat, Joint, Minor, or Student

04

Account Number

Auto-generated 9-digit random number; initial balance = ₹0

Customer Login Portal

After login, customers access 7 services:

Account Holder Detail

View personal KYC info

Account Details

View account number, type & balance

Transaction

Deposit or withdraw funds

Check Balance

Instant balance display

Delete Account

Secure deletion with password confirmation

Money Transfer

Transfer funds via account number

Transactions: Deposit & Withdraw

Deposit

Withdraw

Both operations update the `transactions` and `accountdetails` tables simultaneously. Withdrawals include a safety check — if balance is insufficient, the transaction is rejected with a warning.

Money Transfer Feature

How It Works

Step 1

Enter sender & receiver account numbers

Step 2

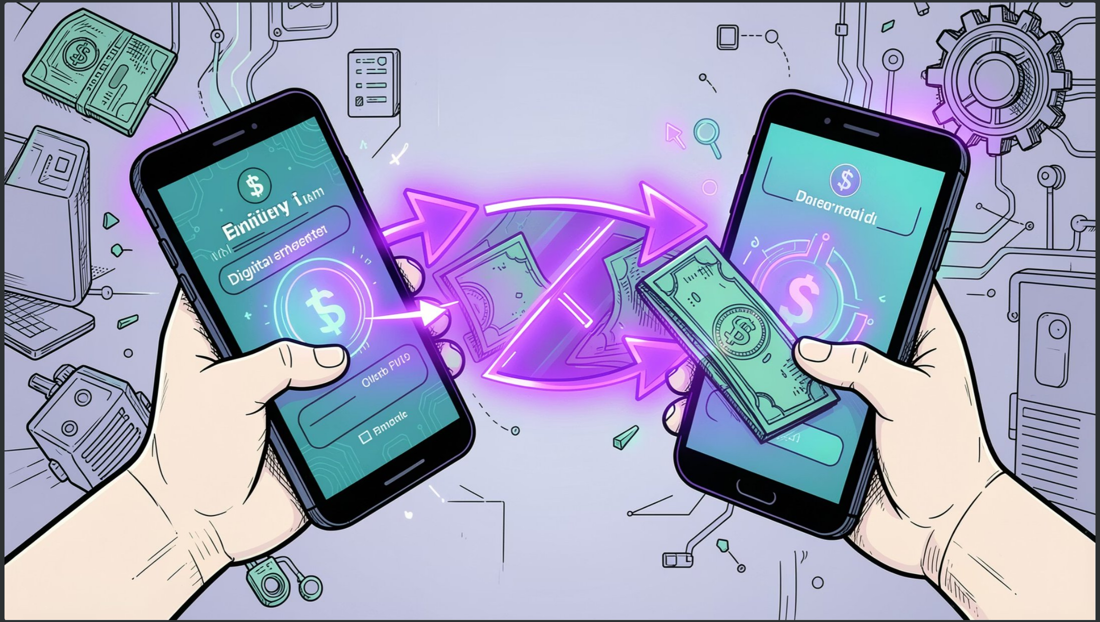
Validate both accounts exist

Step 3

Check sender has sufficient balance

Step 4

Debit sender, credit receiver atomically



Admin Panel

Secured with credentials `infinty88 / user@2204`. Admins can:

👁 View All Customers

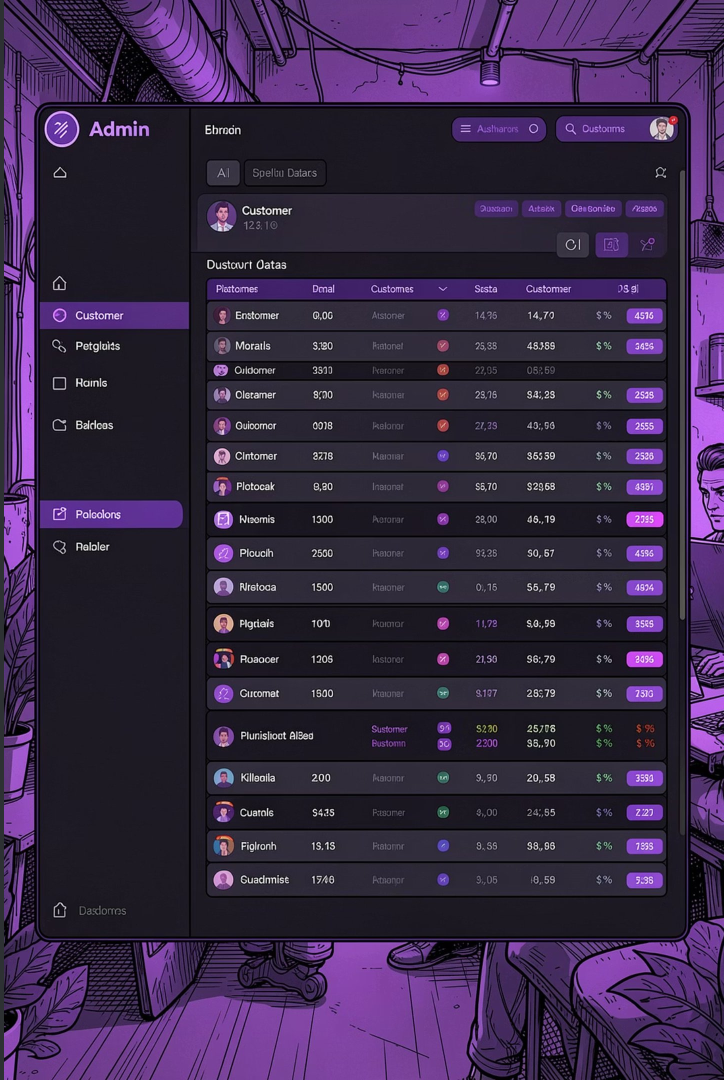
Displays joined `accountholder` + `accountdetails` data in a formatted table

🔍 Search Customer

Lookup by account number; shows both account and personal details

🗑 Delete Customer

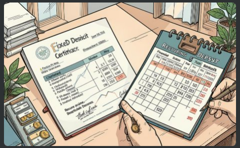
Confirms before removing account records from the database



Account Types Supported

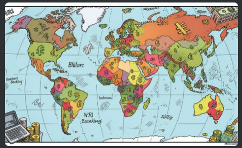
Savings & Current

Standard accounts for everyday banking needs



FD & RD

Fixed and Recurring Deposit accounts for investments



NRI & Salary

Tailored for non-resident Indians and salaried employees

Demat, Minor & Student

Specialized accounts for trading, minors, and students

Key Takeaways

10

Account Types

Savings to Student accounts supported

7

Customer Services

From balance check to money transfer

4

Database Tables

Fully relational MySQL schema

🕵 **Infinite Wallet Bank** demonstrates end-to-end banking logic using Python + MySQL — from secure login and KYC storage to real-time transactions and admin oversight.

Future Enhancements

Our vision extends beyond the current capabilities, focusing on innovation, security, and user experience.

Mobile App Integration

Native iOS/Android applications for on-the-go banking and instant notifications.

Advanced Security Features

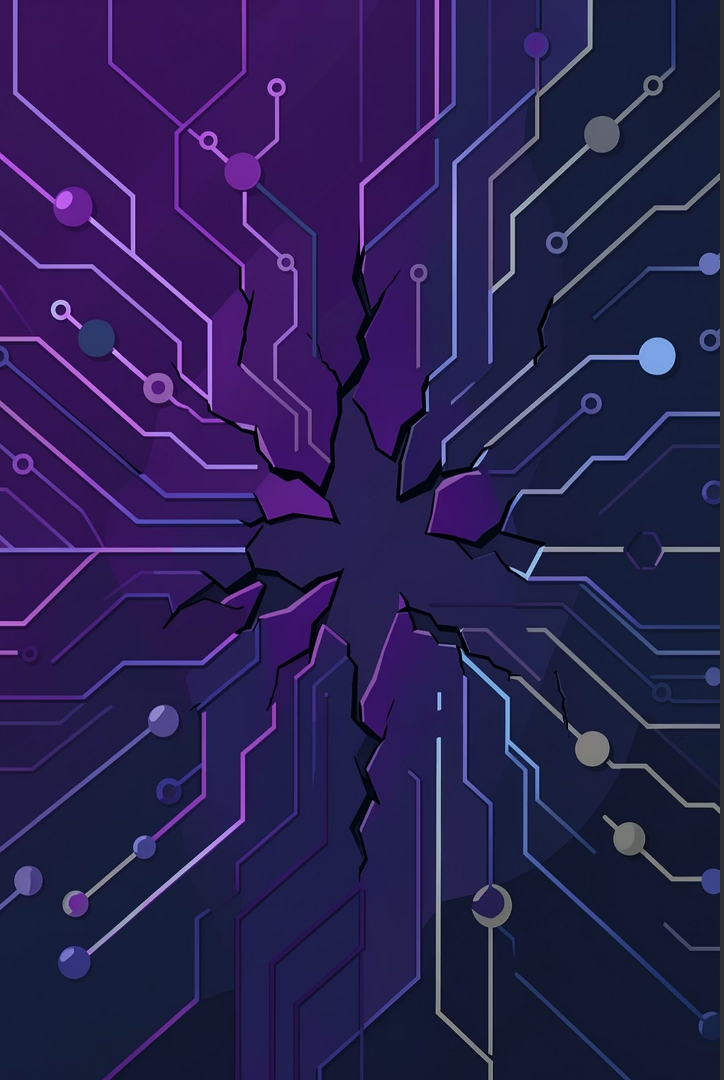
Implement two-factor authentication (2FA) and biometric login for enhanced account protection.

AI-Powered Financial Insights

Integrate AI for personalized spending analysis, budgeting tools, and fraud detection.

Third-Party API Integration

Open API access for seamless connection with external financial services and fintech partners.



Project Limitations

While robust for its scope, the current project has areas for growth when compared to full-scale banking systems.

Command-Line Only Interface

Currently, the banking system operates solely via a command-line interface, lacking a modern graphical user experience for customers.

Basic Security Features

Advanced security measures like two-factor authentication (2FA) or biometric login are not yet implemented.

Limited Scalability

The architecture is primarily designed for smaller user bases and may face performance challenges with high transaction volumes.

Fundamental Auditing & Reporting

Comprehensive audit trails and advanced reporting functionalities for compliance and in-depth analysis are not extensively developed.

Project Strengths

Despite current limitations, the Infinite Wallet Bank project offers several foundational advantages.

Robust Core Functionality

Successfully implements critical banking operations: account creation, secure transactions, and money transfers.

Relational Database Design

Utilizes a structured MySQL database ensuring data integrity and efficient management of customer, account, and transaction data.

Clear Admin Oversight

The admin panel provides essential tools for customer management, enhancing operational control and support capabilities.

Scalable Foundation

Built on Python and MySQL, offering a flexible and well-understood architecture for future enhancements and feature additions.

Thank You!

We appreciate your time and interest in the Infinite Wallet Bank project.

Together, we build infinite trust and infinite growth.